

Systrophē II: Quantum Field Theory on a Tipler-Pair Background

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Abstract

We extend the open-source SYSTROPHE package (github.com/Zynerji/systrophe) from a classical-GR Tipler-pair simulator (v0.1–v0.6) to a quantum-field-theoretic laboratory (v0.7–v0.13). The new infrastructure supports: (i) renormalised stress-energy tensor $\langle T_{\mu\nu} \rangle$ on the Lewis–Papapetrou exterior via point-splitting and the Hadamard parametrix, with exact recovery of the conformal anomaly $K_{\text{Kretsch}}/(2880\pi^2)$ in vacuum; (ii) closed-form Atiyah–Patodi–Singer η -invariant on the $\mathbb{Z}_3$ Möbius cover with verified $\eta_0 + \eta_1 + \eta_2 = 0$ anomaly closure and Callan–Harvey inflow balance; (iii) joint Floquet quasi-energies on the (t, b) Brillouin product torus with $\mathbb{Z}_3$ cyclic-permutation symmetry; (iv) Unruh acoustic-metric mapping ($c^2 - v^2 = F$) identifying the chronology horizon as an acoustic horizon, with the gravitational and acoustic Hawking temperatures coinciding to machine precision; (v) Brown–Maclay $\langle T_{\mu\nu} \rangle$ at a Casimir cavity in flat space with LP curvature correction; (vi) Newton–Kantorovich back-reaction solver demonstrating the linear-walk failure of naïve Picard iteration. The package now contains 406 passing tests and ships the math via a clean module-per-construction layout. We document the open speculative items as ansatz-level interpretations needing external input (cavity geometry, vacuum-selection criterion, etc.), and propose specific experimental tests in BEC sonic-horizon apparatuses.

1 Introduction

SYSTROPHE v0.1 implemented the closed-form van Stockum / Tipler exterior with log-periodic structure and a co-axial pair construction allowing tunable CTC interference. The first whitepaper [1] established the classical-GR mathematics: exact $\alpha = \sqrt{4a^2 - 1}$, $F(r) \propto \cos(\alpha \ln r/R + \delta)$ sinusoid, phasor-sum closure, and the structural $\delta = \pi$ extinction. The present paper extends to QFT on this background.

1.1 Scope of Systrophē II

The v0.7–v0.13 series of releases adds the following independent machinery, each with a clean module and a separate test suite:

- **v0.7:** Radial Dirac operator on the LP exterior with bound-state spectrum (`dirac.py`, `dirac_spectrum.py`).
- **v0.8:** Quantum-field-theory in curved spacetime infrastructure: Dirac sea pressure, particle creation at the chronology horizon, QFTCS back-reaction trace, vacuum-energy proxy (`dirac_sea.py`, `particle_creation.py`, `qftcs_backreaction.py`, `quantum_diagnostics.py`).
- **v0.9:** Point-splitting renormalisation; full Riemann tensor, Kretschmann scalar, exact 4D conformal-anomaly trace, and DeWitt heat-kernel a_2 coefficient (`point_splitting.py`).
- **v0.11:** Research-grade adiabatic Floquet on the radial Dirac problem (`floquet.py`).

- **v0.12:** Topological-Casimir mode sums on the $\mathbb{Z}_3$ cover with the Hurwitz ζ at $s = -3$ closed form (`casimir.py`).
- **v0.13:** *This paper.* Eight new modules covering the post-anomaly machinery (Sections 2–5 below): `hadamard_offtrace`, `anomaly_inflow`, `floquet_mobius`, `acoustic_metric`, `casimir_throat`, `newton_kantorovich`, `tipler_fractal`, `horned_torus`; plus four extension modules `back_reaction`, `floquet_engineering`, `dsi_observables`, `adm_export`, `d_ctc`.

2 Renormalised stress tensor and off-trace components

The package’s exact 4D conformal anomaly is

$$\langle T^\mu_\mu \rangle_{\text{ren}} = \frac{K_{\text{Kretsch}}}{2880 \pi^2} \quad (\text{vacuum, } R_{\mu\nu} = 0). \quad (1)$$

Module `hadamard_offtrace.py` extends this to the full *tensor*

$$\langle T_{\mu\nu} \rangle_{\text{ren}} = \frac{1}{2880 \pi^2} R_{\mu\rho\sigma\tau} R_\nu^{\rho\sigma\tau}, \quad (2)$$

whose trace recovers (1) exactly. The state-independent geometric piece (2) is testable against the trace anomaly numerically; state-dependent corrections (Boulware vs. Hartle–Hawking) are documented but not implemented (they require mode-sum machinery beyond this paper).

We verify the trace-matching condition numerically:

```
trace_decomp = trace_decomposition(vs, r=2.0)
trace_decomp["trace_anomaly_check"] # < 1e-13
```

3 Anomaly inflow on the $\mathbb{Z}_3$ cover

The angular Dirac operator on S^1 with twist α has APS η -invariant

$$\eta(\alpha) = 1 - 2\alpha, \quad \alpha \in (0, 1), \quad (3)$$

with $\eta(0) = 0$ (symmetric regularisation).

For the $\mathbb{Z}_3$ Möbius cover ($\alpha_b = b/3 + \gamma_{\text{eff}}/(2\pi)$), the three branches have

$$(\eta_0, \eta_1, \eta_2)|_{\gamma=0} = (0, \frac{1}{3}, -\frac{1}{3}), \quad (4)$$

and

$$\sum_{b=0}^2 \eta_b = 0, \quad (5)$$

the anomaly closure condition. A non-zero gauge twist γ_{eff} breaks (5); the residual is cancelled by a Chern–Simons bulk inflow with coefficient $k_{\text{CS}} = 1/(24\pi^2)$:

$$k_{\text{CS}} \cdot q_{\text{flux}} = -\frac{1}{2} \sum_b \eta_b(\gamma_{\text{eff}}). \quad (6)$$

This is the Callan–Harvey relation evaluated explicitly on the $\mathbb{Z}_3$ cover; verified by direct construction in `z3_anomaly_inflow_balance`.

4 Acoustic-metric mapping and analog Hawking

Unruh’s 1981 acoustic-metric mapping identifies the LP angular metric as an acoustic line element via

$$\rho_{\text{acoustic}} = \sqrt{L}, \quad v_\phi = K/\sqrt{L}, \quad c^2 - v_\phi^2 = F. \quad (7)$$

The chronology horizon $F = 0$ coincides exactly with an *acoustic* horizon ($c = v$); the CTC region ($F < 0$) is the *supersonic* flow region. The acoustic Hawking temperature

$$T_H^{\text{ac}} = \kappa_{\text{ac}}/(2\pi), \quad \kappa_{\text{ac}} = \frac{1}{2}|dF/dr|_{r=r_H} \quad (8)$$

equals the gravitational Hawking temperature at the LP chronology horizon to machine precision (relative difference $< 10^{-12}$; `compare_acoustic_vs_gravitational_T_H`).

Experimental access. The acoustic-analog mapping is the *only currently lab-tractable path* to the Systrophē physics. In a Bose–Einstein condensate (BEC) with a quantum vortex of winding ℓ , the azimuthal flow $v_\phi(r) = \hbar/(mr)$ automatically violates $v_\phi(R) > c/2$ once $R < 2\xi_{\text{BEC}}$ (the healing length), reproducing the Tipler supercritical condition $\omega R > 1/2$. A $\mathbb{Z}_3$ triple-vortex configuration realises the Möbius cover. The full design proposal is in `docs/EXPERIMENTAL_ACOUSTIC_AN`.

5 Floquet engineering and discrete-scale invariance

5.1 Joint Floquet on (t, b)

Module `floquet_mobius` computes Floquet quasi-energies on the joint time-circle $\times \mathbb{Z}_3$ branch space. The three-level Hamiltonian

$$H_{\text{static}} = \text{diag}(e_0, e_1, e_2) + g \cdot (|0\rangle\langle 1| + |1\rangle\langle 2| + |2\rangle\langle 0| + h.c.) \quad (9)$$

is driven by $V(t) = A \sin(\Omega t)(S + S^\dagger)$, where S is the $\mathbb{Z}_3$ cycle-shift. Quasi-energies satisfy $\mathbb{Z}_3$ cyclic-permutation symmetry; resonances open at $\Omega = e_b - e_{b'}$.

The `floquet_engineering` extension module maps the (amplitude, frequency) plane and identifies the CTC stability gap

$$\Delta_{\text{stable}}(\text{amp}, \Omega) = \min_{i \neq j} |\epsilon_i - \epsilon_j|/\Omega, \quad (10)$$

quantifying drive-induced stabilisation or destabilisation of CTC bands.

5.2 Tipler-sinusoid discrete-scale invariance and cascade-DSI

The Tipler sinusoid $F(r) = \cos(\alpha \ln r/R + \delta)$ has zero set $\{r_n = r_0 e^{\pi n/\alpha}\}$: a geometric progression with ratio

$$\rho = e^{\pi/\alpha}, \quad (11)$$

verified to machine precision in `base_tipler_sinusoid_is_dsi`. This is the textbook DSI signature of a log-periodic precursor (Sornette 1998). A multi-cylinder cascade

$$F_{\text{cascade}}(r) = \sum_{k=0}^{L-1} A_k \cos(\alpha_k \ln r/R + \delta_k), \quad \alpha_k = \alpha_0 \lambda^k, \quad (12)$$

extends to non-trivial box-counting dimension (measured > 0.3 in `cascade_box_dimension`). `dsi_observables` ports this machinery to data analysis: log-periodic precursor fits and Lomb–Scargle log-frequency search.

6 Back-reaction and self-consistency

The `back_reaction` module wires `hadamard_offtrace` and `newton_kantorovich` into a composite back-reaction residual

$$\mathcal{R}(\delta) = w_T \sum_r \|\langle T_{\mu\nu} \rangle(s_1; r)\| + w_L \sum_r |L_{\text{pair}}(\delta; r)|. \quad (13)$$

For the matched pair (identical a, R, A), the residual is minimised at $\delta = \pi$, matching the v0.11 CTC log-measure result. This is *not* a full Einstein-coupled back-reaction solve (the spatial metric is not updated between iterations); it is a static residual minimisation. Full back-reaction requires an ADM-coupled solver, which is enabled by the `adm_export` module providing initial-data hand-off for Einstein Toolkit / similar NR codes.

7 D-CTC and quantum information

Module `d_ctc` implements Deutsch’s 1991 CTC fixed-point equation

$$\rho_{\text{CTC}} = \text{Tr}_{\text{CR}}[U(\sigma_{\text{CR}} \otimes \rho_{\text{CTC}})U^\dagger] \quad (14)$$

on the $\mathbb{Z}_3$ branch space as the natural 3-dimensional CTC register. The cyclic-shift unitary $U = R_{\text{CR}} \otimes S$ admits the maximally-mixed CTC state as a Banach fixed point; the iteration converges to machine precision in tens of iterations.

This puts SYSTROPHE in conversation with the Aaronson–Watrous (2008) PSPACE result on D-CTC complexity: the underlying spacetime geometry is here concretely instantiated.

8 Open speculative items

Six ansatz-level claims from the original Grok-chat analysis remain not directly testable from inside the codebase and are documented in `docs/INTERPRETATIONS.md`:

1. $\mathbb{Z}_3$ cover physically realises a wormhole throat (needs gluing map);
2. Casimir negative energy replaces exotic matter (needs cavity geometry);
3. Self-consistent δ -iteration selects the chronology-protecting phase (NK solver ready; needs $F(\delta)$);
4. Möbius monodromy equals a $U(1)$ Wilson loop (needs gauge field);
5. DCE off-resonance photon flux (needs cavity Q + modulation);
6. Throat cavity is the natural QFT vacuum (needs vacuum-selection criterion).

Each item is an open research project given external input.

9 Summary and outlook

SYSTROPHE v0.13 provides the smallest open-source toolkit known to the author that (i) constructs an exact closed-form CTC spacetime, (ii) computes renormalised quantum-field-theoretic observables on it, and (iii) supports both an experimental analog (BEC vortices) and a quantum-information substrate (D-CTC).

Tests. The package now contains 406 passing tests covering all numerical claims in this paper. The full source is at github.com/Zynerji/systrophe; license is MIT.

Next steps. (a) BEC-vortex experimental design collaboration (CONTACTS.md, Tier 1); (b) full back-reaction NR run via ADM hand-off; (c) extension to the multi-cylinder cascade DSI as a model of analog-cosmological log-periodic precursors.

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